

EDUCATION

Courant Institute of Mathematical Sciences, New York University , computer science <i>PhD.</i>	starting 2026.9
• Verifiable systems and theoretical computer science.	
Stanford University , computer science <i>M.S.</i> and mathematics <i>B.A.</i>	2019.9 - 2024.6
• General Yaowu Wang Scholarship.	
St Paul's School	2017.9 - 2019.6
• Full scholarship.	

EXPERIENCE

a16z	Researcher, 2025.4 - Present
• Design and build Jolt , an open-source zero-knowledge virtual-machine in Rust.	
• Systems performance, sumcheck algorithm optimization.	
• Proof compression research, designed torus compression system, gave theoretical optimality proof and reframed method conceptually via Hilbert 90.	
• Turned folklore results into rigorous proofs , audited security in elliptic curve optimizations.	
Millennium	Researcher, 2024.6 - 2025.4
• Convex optimization, robust statistical estimation.	
Stellar Development Foundation	Consultant, 2023.6 - 2023.9
• Byzantine fault tolerance, C++, stellar-core .	
• Performance optimization. Improved maximum transaction rate per second by 17% via changes to the consensus protocol which reduce number of communication rounds in overlay network. Formal verification in TLA+.	
Distributed Systems and TCS Research, Stanford University	Researcher, 2023.9 - 2024.6
• Under David Mazieres, design and implement a general open membership Byzantine fault tolerance consensus protocol framework and implement applications (e.g. certificate authority management and automatic password recovery with hardware security modules) on top of it. Github repo .	
• Research PCP under Aviad Rubinfeld. Read proof of the 2-to-2 Games Conjecture. Proved a result about Convex Continuous Class Complexity.	
Topology Research, Stanford University	Researcher, 2020.7 - 2022.5
• Research under Ciprian Manolescu. Floer theory, Seiberg-Witten, knot invariants. Increased the number of possible counterexamples for the smooth Poincare conjecture from 21 to 22 by detecting RBG links that satisfy the conditions outlined in this paper .	
• Honours thesis under Mohammed Abouzaid. Proved a result sketched by Quillen relating the Gysin homomorphism of complex cobordism to the Grothendieck residue symbol. Produced a recursive formula for the correction term in generalized complex-oriented cohomology valued Gromov-Witten theory equipped with a quantum product.	

SKILLS

Computer Science: Rust, C/C++, Python, formal verification, convex optimization, distributed systems, cryptographic protocol design, performance engineering.
Mathematics: number theory, algebraic geometry, symplectic topology, Floer theory, gauge theory, concentration inequalities.